MILITARY SPECIFICATION

BRUSHES, FLAP, ROTARY, ABRASIVE, GENERAL SPECIFICATION FOR

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification establishes the requirements for nylon-aluminum oxide brushes (see 6.1). It also establishes the requirements for a mandrel used to adapt the brushes for use in a high speed drill.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Cadmium Plating, Electro Deposited
QQ-S-634	Steel Bar, Carbon, Cold Finished, (Standard Quality)
QQ-S-698	Steel, Sheet and Strip, Low Carbon

Military

MIL-A-3816	Abrasives and Abrasive Products for Shipment and Storage, Packaging and Packing of
MIL-A-9962	Abrasive Mats, Non-Woven, Non-Metallic
MIL-B-81949/1	Brushes, Flap, Rotary Abrasive
MIL-B-81949/2	Brush Mandrel, Flap, Rotary, Abrasive

FSC 5345

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-143	Specifications and Standards, Order of Procedure for the Selection of

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specification sheets - The detail requirements applicable to the products (brushes and mandrel) shall be as specified herein and in accordance with applicable specification sheets. In the event of conflict between the requirements of this specification and the specification sheet, the requirements of the specification sheet shall prevail.

3.2 First article - The products furnished under this specification shall have passed the first article inspection specified herein(see 4.3).

3.3 Selection of Specifications and Standards - Specifications and standards shall be selected in accordance with MIL-STD-143.

3.4 Design -

3.4.1 Brushes- The brushes shall be designed to operate safely at speeds up to 3,800 RPM and to remove paint and surface corrosion from aircraft surfaces.

3.5 Construction - The brush shall be constructed in a manner that will insure an even distribution of flaps. The flaps shall extend from the perimeter of the hub perpendicular to the axis of the hub. They shall be fastened securely to the hub with a thermosetting adhesive that will preclude the flaps from coming loose due to flexure or centrifugal force.

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3.6 Workmanship - The products shall be uniform in quality and free from defects that will be detrimental to the reliability or serviceability of the products. The general appearance of the products shall indicate the highest quality workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection terms - MIL-STD-109 shall govern the definition of terms used herein.

4.2 Classification of inspections - The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3) - First article inspection consists of examinations and tests performed on samples (which are representative of the production item) after the award of a contract. The examinations and tests are conducted to determine that the products meet the requirements of this specification.

(b) Quality conformance inspection (see 4.4) - Quality conformance inspection consists of examinations and tests performed on the production items to determine that the products meet the requirements of this specification.

4.3 First article inspection - First article inspection shall consist of all the inspections of this specification.

4.3.1 First article samples - Unless otherwise specified, as soon as practicable after award of a contract or order, the manufacturer shall submit samples as specified in the contract or order. The samples shall be representative of the construction, workmanship, and materials to be used during production.

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Quality conformance inspection - The quality conformance inspection shall consist of the examination (4.5.1) and the examination of preparation for delivery (4.5.1.3).

4.4.1 Sampling for tests and examination - Sampling shall be in accordance with MIL-STD-105, inspection level III, acceptable quality level 1.5.

4.5 Inspection methods -

4.5.1 Examination - The products shall be thoroughly examined to determine compliance with the requirements of this specification and the applicable specification sheet.

4.5.1.1 Mandrel - The mandrel shall be examined to determine compliance with the requirements specified in MIL-B-81949/2.

4.5.1.2 Brushes - The brushes shall be examined for the defects specified in Table I.

TABLE I

Examine	Defect
Design, Construction and Workmanship	Flaps not set as specified or not evenly distributed. Any loosely packed or incompletely filled brushes Any imperfection which may affect appearance or serviceability. Flaps not evenly or uniformly trimmed.
Materials	Flap material not as specified. Hub not as specified.
Identification marking	Missing, incomplete. Not legible.
Dimensions	Not as specified in MIL-B-81949/1(AS).
Label	Not adhering to the brush, marking not legible, missing, incomplete.

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4.5.1.3 Examination of preparation for delivery - The preservation, packaging and packing of the brushes shall be examined to ascertain that the preparation for delivery conforms to Section 5 of this specification.

4.5.2 Maximum RPM test - The maximum RPM test shall be conducted as follows:

(a) Secure the brush in the mandrel.

(b) Position the mandrel in a high speed pneumatic or electric drill.

(c) Rotate at 3,800 RPM.

(d) Use the brush to remove small areas of corrosion (or paint) from a sheet of aluminum alloy.

(e) Continue step (d) for thirty minutes.

(f) Remove the brush from the mandrel.

There shall be no evidence of the flaps separating from the hub. In addition, there shall be no evidence of the label separating from the side of the brush.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing - Preservation, packaging, and packing shall be in accordance with MIL-A-3816.

5.2 Marking - In addition to any marking specified in the contract or order, interior and exterior shipping containers shall be marked in accordance with MIL-A-3816.

6. NOTES

6.1 Intended use - The flap brushes covered by this specification are intended for use in removing small areas of paint, oxides and other corrosion from aircraft surfaces.

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6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Flap brush grade (See 1.2.1)

(c) Mandrel (if required).

(d) Whether a first article sample is required (See 6.2.1)

(e) Level of packaging and packing required (See 5.1)

6.2.1 Contracts or orders should specify the following provisions for first article inspection.

6.2.1.1 Whether first article inspection is required - When a contractor is in continuous production of the packings from contract to contract, consideration should be given to waive the first article inspections. If inspection is required, indicate:

(a) where the first article inspection is to be conducted (at the contractor's plant or Government or commercial laboratory).

(b) that the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Data - The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order.

6.3.1 First article data - When first article samples are submitted (see 6.2), they should be accompanied by a complete inspection report showing the results of the contractor's inspections.

Preparing Activity:

Navy - AS

(Project No. 5345-N038)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-B-81949(AS) BRUSHES, FLAP, ROTARY, ABRASIVE, GENERAL SPECIFICATION FOR

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

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